

Histograms Worksheet Answers

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1 Useful questions asked:

1. **Does the outlier count as a mode or hill?** No, it doesn't. The idea with a mode is that you want a clumping of points together, usually showing a population. An outlier (a single point) doesn't fit that definition. IF you have a clumping of multiple outliers together then they aren't outliers but some type of subpopulation (and possibly a another mode) but that is a rare case. A single point won't count as a mode.
2. **To be bimodal does the second hill have to be as large as the first hill?** Also no. It's not that the two subpopulations have the same height/counts but that there is some form of distinguishable groupings. It's alright if one hill is relatively small. For example, the bottom right histogram could have the second hill only a fourth as large as the first hill and I'd still be very comfortable calling it bimodal.
3. **What happens if I think the histogram is neither skewed nor symmetric?** Then you say that! Not being symmetric (where the histogram can't be folded on itself) and not being skewed (neither side has a tail) does happen and that's okay. Just identify it and explain your thinking.
4. **How do we know if an outlier is "bad"?** My personal rule of thumb (good or bad...) is that if there is only about one empty bin on the histogram I don't mind too much. Part of this is how strongly we want to default towards mean and st. dev. and part of this is that the shape of the histogram can change some depending on how large we make the bins (eg slightly fewer bins may "hide" the outlier).

2 Top Right

- Unimodal

- No Outliers
- Roughly Symmetric
- Great case for mean and st. dev.
- This and the third from top on the right are both normal distributions (the classic bell shape)

3 Top Left

- Unimodal
- Right skewed
- 1 outlier
- 5# Summary (the skew is baaaad)
- This is called an exponential distribution, it's a special type of the Gamma distribution.

4 Both histograms on the second row

- Bimodal
- Neither skewed or symmetric
- No outliers
- 5 # summary (not symmetric enough)

5 Third histogram from the top, on the left

- Either bimodal or uniform
 - The data was generated from a uniform distribution which would have a roughly flat line for the height of the bins in the histogram
 - But there is definitely a valley looking thing which would suggest bimodal (the two modes being the far left and right). This is an acceptable interpretation even if the “truth” is that it's uniform. Randomness isn't always your friend and can produce meaningless patterns.

- Symmetrical
- No outliers
- Mean and St. Dev. (not skewed nor bad outliers)

6 Third histogram from the top, on the right

- Unimodal
- Symmetric
 - I do see where some people are hesitant with this given that the histogram isn't perfectly lining up if folded in on itself.
 - I'd still lean symmetric enough
- 1 outlier on the left
- Mean and St. Dev. (the outlier isn't bad and it's symmetric enough)

7 Bottom left

- Bimodal
- Symmetric
- No Outliers
- Mean and St Dev are great for this

8 Bottom Right

- Unimodal
- 1 outlier
- Non symmetric
- 5 # Summary for this
- This is what's called a "Gamma" distribution.